



**Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)**

Experiment 7

QUANTITATION OF PROTEINS

Pre-Lab Questions:

- 1) What is the concentration of unknown sample have an absorbance of 0.198 and the absorbance of 4 g/L standard is 0.356 ?
- 2) Why the concentration of Trichloroacetic acid is very low?
- 3) What is the blank reagent in turbidimetry test?

Precautions

- Biuret reagent is a strong base ($\text{pH} > 12$) and toxic due to the copper. Whenever you work with Biuret Reagent solutions you must: wear gloves and clean up any spills.
- Follow directions for proper disposal.
- The protein standard solutions are harmless due to containing albumen.

Objectives:

- 1-Determination of Protein Concentration in terms of mg/dL
- 2-To assay protein samples of unknown concentration
- 3-Analyze data from standard curve and unknowns
- 4-Learn the principles of protein assays.
- 5-Explore the methods used for quantification of proteins.
- 6-Study the effect of common laboratory reagents on the methods of for protein quantification.

Introduction

The measurement of protein concentration in an aqueous sample is an important assay in biochemistry research and development laboratories for applications ranging from enzymatic studies to providing data for biopharmaceutical lot release. Spectrophotometric determination of protein concentration assays are methods that use UV and visible spectroscopy for rapid and accurate results relative to a standard. These assay methods is simple, reproducible and tolerant to many interfering substances. It can be run at a high throughput, using inexpensive reagents with equipment found in the majority of biochemical laboratories.

The most important two methods of quantitation of proteins are :

A) Turbidimetry:

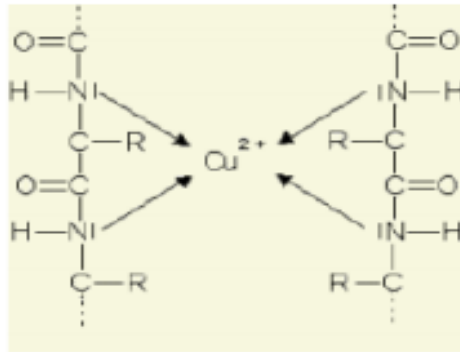
Turbidimetry (the name being derived from turbidity) is the process of measuring the loss of intensity of transmitted light due to the scattering effect of particles suspended in it. Light is passed through a cuvette containing a solution. A photoelectric cell collects the light which passes through the cuvette. A measurement is then given for the amount of absorbed light.

A turbidimetric method for protein quantitation involves the use of trichloroacetic acid as the precipitating agent. Maximal turbidity develops in less than 30 mins and is stable for at least 120 min. A relationship between turbidity at 620 nm and protein concentration is linear up to 200 mg/dl protein.

This procedure compared favorably with the most widely used protein quantitation methods in simplicity, sensitivity, and specificity.

B) The Biuret Assay:

The Biuret reaction involves a reagent containing copper (cupric) ions in alkaline solution. Molecules containing 2 or more peptide bonds associate with the cupric ions to form a coordination complex that imparts a purple color to the solution with $\lambda_{\text{max}} = 540 \text{ nm}$ (figure 1) . The purple color of the complex can be measured independently of the blue color of the reagent itself with a spectrophotometer. The amount of purple color that forms is directly proportional to the quantity of protein in your samples. For this method, the relationship between absorbance and concentration is linear up to 800 mg/dL.



Figur1: The Biuret method is specific, easy to carry out and takes little time for protein estimation

Experimental procedure

Part A Turbidimetry

Materials:

- 7- 3% Trichloroacetic acid or sulfosalicylic acid
- 8- A protein set of standards (60, 120, and 180 mg/dL)

Procedure

- 1- Prepare a blank solution of 2.0 mL acidic reagent with 0.5 mL distilled water.
- 2- Mix 0.5 mL of each of the standards with 2.0 mL of trichloroacetic acid or sulfosalicylic acid and read their absorbance after 10 minutes at 620 nm.
- 3- Do the same for the unknowns.
- 4- Tabulate your results and construct a calibration curve from which the concentration of the protein can be determined.

Part B The Biuret Assay

Materials:

- 1- A set of protein standards (250, 500, and 750 mg/dL, freshly prepared Albumin).

- 2- Biuret Reagent (Dissolve 3 grams of copper sulfate and 9 grams of sodium potassium tartrate and 5 grams of potassium iodide in 1000 mL of 0.2 M sodium hydroxide.
- 3- Water bath at 37 °C

Procedure

- 1- Prepare a blank solution of 3.0 mL of Biuret reagent with 2.0 mL distilled water.
- 2- Add 3.0 mL of Biuret reagent to 2.0 mL of each of the standards and to the unknown protein solution.
- 3- Shake well and keep in water bath for 10 minutes at 37 °C.
- 4- Read absorbance for standard solution and unknown solution at 540 nm.
- 5- Tabulate your results and draw the relationship between absorbance and concentration to construct a calibration curve.
- 6- Determine the concentration of the protein in the unknown sample.

References:

1. "A Rapid and Sensitive Method for the Quantitation of Microgram Quantities of Protein Utilizing the Principle of Protein-Dye Binding", MARION M. BRADFORD
Reproduction Research Laboratories, Department of Biochemistry, University of Georgia, Athens, Georgia 30602 Received September 11, 1975; accepted January 29, 1976
2. "Quantitation of Protein" Chapter eight, James E. Noble and Marc J. A. Bailey
3. *Mary C. Haven; Gregory A. Tetrault; Jerald R. Schenken (1994). Laboratory Instrumentation. John Wiley and Sons.* ISBN.
4. "A turbidimetric determination of protein by trichloroacetic acid" , Wahn Soo Choi ,Kae Jong Chung ,Man Sik Chang ,Jae Kwang Chun, Hyang Woo Lee and Sung Youl Hong
5. "Biuret Protein Assay" ,Revised Fall 2010



**Faculty of Science and Arts
Applied Chemistry
Biochemistry Lab (CHEM 266)
Experiment 7**

Quantitation of Proteins

Report Author:

Lab Partner:

Date Due:

Purpose:

Data and Results :

A: Turbidimetry

Test solution	Concentration (mg/dL)	Absorbance
1	60	
2	120	
3	180	
Unknown #		

A: Biuret test

Test solution	Concentration (mg/dL)	Absorbance
1	250	
2	500	
3	750	
Unknown #		

-On a graph paper plot absorbance (A) versus concentration (C).

Conclusion:

Errors and problems you encountered and their possible remedies: